MIA GAIA POLANSKY, PH.D.

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I am an applied researcher who builds principled, efficient models that bridge classical geometric reasoning and modern deep learning. My research at Harvard and Google focused on developing visual learning systems (probabilistic 3D Gaussian splatting and geometry-aware attention networks) that train robustly under constraints like limited labels, noisy data, or tight memory budgets. I am drawn to problems where explicit structure improves the reliability and generalization of otherwise data-driven approaches, and I enjoy the practical work of getting there in practice: diagnosing instability, improving convergence, and building infrastructure that makes training faster and more reliable.

EDUCATION

Harvard University | Cambridge, MA — *February 2026*

Ph.D., Engineering Sciences — Advised by Prof. Todd Zickler

Rice University | Houston, TX — *May 2017*

B.S., Electrical Engineering — Data Science Specialization

RESEARCH & INDUSTRY EXPERIENCE

Graduate Researcher | Harvard University | Cambridge, MA — *2017 – 2026*

Probabilistic 3D Scene Reconstruction & Differentiable Rendering | euleriansplatting.github.io

- Developed a probabilistic Gaussian Splatting framework that uses a compact “hashed probability pyramid” data structure to learn high-resolution 3D density fields, replacing hand-tuned density-control heuristics with scalable end-to-end optimization and reducing memory requirements by over 800× compared with a dense grid
- Achieved state-of-the-art novel-view synthesis PSNR, showing that the model could recover high-quality visual structure from image supervision without COLMAP-based initialization
- Derived and implemented a control-variate gradient estimator that stabilized stochastic training, reduced gradient variance by orders of magnitude, and improved convergence from random initialization
- Modified a PyTorch differentiable renderer’s backward pass to recover per-primitive image contributions without extra forward renders, preserving efficiency while producing a cleaner training signal
- First-author publication (CVPR 2026): "Eulerian Gaussian Splatting using Hashed Probability Pyramids”

Student Researcher | Google Research | Cambridge, MA — *2022 – 2024*

Geometry-Aware Attention & Structured Visual Representation | boundaryattention.github.io

- Developed Boundary Attention, a novel geometry-aware local attention mechanism that encodes geometric continuity biases, enabling a compact model to recover fine boundary structure from noisy images while remaining robust and parameter efficient
- Built a synthetic-data training pipeline that allowed the model to learn useful visual structure from simple geometric data and transfer to noisy real-world photographs without fine-tuning or expensive real-image labels
- Matched or outperformed diffusion-based boundary detectors over 1000× larger, showing how targeted model design and domain-specific inductive biases can reduce the need for scale alone
- Used ablations and learned-representation visualizations to diagnose model behavior and understand why adaptive windowing, iterative refinement, and parameter sharing improved consistency, robustness, and convergence
- First-author publication (ECCV-W 2024): "Boundary Attention: Learning Curves, Corners, Junctions and Grouping”

Undergraduate Research Assistant | Rice Efficient Computing Group | Houston, TX — *Summer 2015*

- Modeled analog noise effects on deep neural networks (GoogLeNet, AlexNet) in Python and C, characterizing degradation under hardware-realistic conditions
- Publication (ISCA 2016): "RedEye: Analog ConvNet Image Sensor Architecture for Continuous Mobile Vision"

TEACHING & MENTORSHIP

Teaching Fellow | Harvard University — *2018 – 2019*

CS 283: Computer Vision, CS 175: Intro to Computer Graphics, BE 128: Medical Imaging* *Distinction in Teaching

Head Fellow for Harvard Diversity, Inclusion, and Belonging (DIB) Initiative | Harvard University — *2021 - 2022*

- Developed a graduate mentorship rotational program that pairs Harvard undergraduates with graduate mentors to increase research accessibility

Harvard Leverett House Academic & Career Advisor | 2018 – 2022

- Provided career and academic advising for sophomore Harvard engineering undergraduates

Lumiere Education Mentor *2020-2022*

- Mentored high school students through research projects related to computer vision or machine learning

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Core Areas: Attention Mechanisms & Transformer Architectures (ViT, local attention), Diffusion Models, Geometric Inductive Biases, 3D Scene Reconstruction, Probabilistic Modeling & Gradient Estimation

Languages & Tools: Python (PyTorch, JAX), C++, Slang